

DOC-01

# User Roles & Permissions

Role definitions, module access matrix, and workflow diagrams



|         |                                   |
|---------|-----------------------------------|
| Project | GPA-ERP V5.0                      |
| Type    | Construction Cost Control ERP     |
| Stack   | FastAPI · Next.js 14 · PostgreSQL |
| Version | V5.0                              |

# GPA-ERP — User Roles & Permissions

## Role Definitions

| Role                | Description                                                                              | Typical Person          |
|---------------------|------------------------------------------------------------------------------------------|-------------------------|
| <b>SUPER_ADMIN</b>  | Full system access. Manages users, menus, approval matrix, vault. Can do everything.     | System Administrator    |
| <b>MD</b>           | Managing Director. Final approver on large expenses. Signs legal documents. Confirms AR. | CEO / Managing Director |
| <b>PM</b>           | Project Manager. Manages projects, approves mid-range expenses, signs some legal docs.   | Project Manager         |
| <b>COST_CONTROL</b> | First verifier on all submitted expenses. Manages cost codes, cost centres.              | Cost Control Officer    |
| <b>FINANCE</b>      | Marks approved expenses as paid. Manages financial disbursement.                         | Finance / Accounting    |
| <b>GA</b>           | General Affairs. Manages inventory in/out, creates petty cash reports.                   | GA Officer              |
| <b>STAFF</b>        | Basic user. Creates draft expenses, creates draft legal documents.                       | Any employee            |

## Module Access Matrix

| Module                                | SUPER_ADMIN | MD | PM | COST_CONTROL | FINANCE | GA | STAFF |
|---------------------------------------|-------------|----|----|--------------|---------|----|-------|
| <b>Dashboard</b>                      | ■           | ■  | ■  | ■            | ■       | ■  | ■     |
| <b>Projects — view</b>                | ■           | ■  | ■  | ■            | ■       | ■  | ■     |
| <b>Projects — create/edit/delete</b>  | ■           | ■  | ■  | ■            | ■       | ■  | ■     |
| <b>Projects — bulk import (Excel)</b> | ■           | ■  | ■  | ■            | ■       | ■  | ■     |
| <b>Revenue (AR) — view</b>            | ■           | ■  | ■  | ■            | ■       | ■  | ■     |
| <b>Revenue (AR) — create/edit</b>     | ■           | ■  | ■  | ■            | ■       | ■  | ■     |

| Module                                         | SUPER_A<br>DMIN | MD | PM | COST_CO<br>NTROL | FINANCE | GA | STAFF |
|------------------------------------------------|-----------------|----|----|------------------|---------|----|-------|
| <b>Revenue (AR)</b> — confirm                  | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — create draft                 | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — submit                       | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — verify                       | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — approve (matrix)             | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — mark paid                    | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Expenses</b> — hard lock                    | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Petty Cash</b> — create/manage              | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Legal Docs</b> — create/edit                | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Legal Docs</b> — sign/reject                | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Inventory</b> — view                        | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Inventory</b> — create items / transactions | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Action Center</b> (my queue)                | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Reports</b>                                 | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Vault</b> (cost codes, rules, audit)        | ■               | ■  | ■  | ■                | ■       | ■  | ■     |
| <b>Admin Panel</b> /admin                      | ■               | ■  | ■  | ■                | ■       | ■  | ■     |

| Module            | SUPER_A<br>DMIN | MD | PM | COST_CO<br>NTROL | FINANCE | GA | STAFF |
|-------------------|-----------------|----|----|------------------|---------|----|-------|
| Users —<br>manage | ■               | ■  | ■  | ■                | ■       | ■  | ■     |

## User Role Flowchart

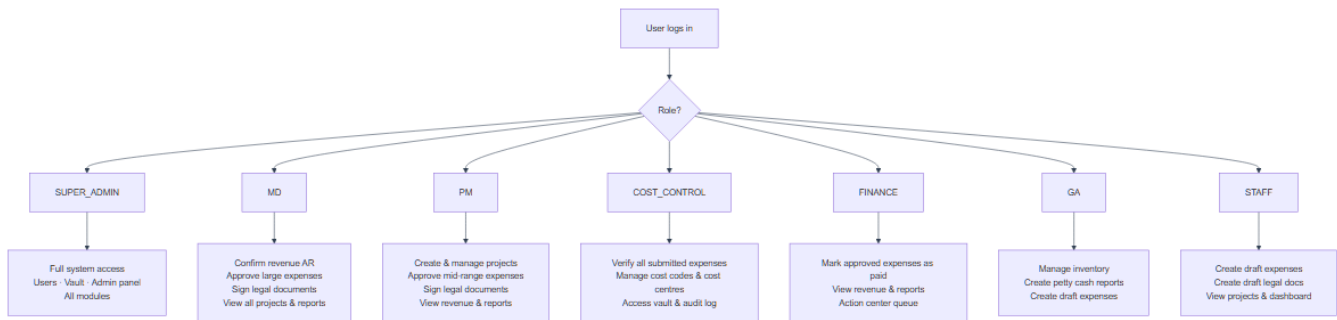


Figure 1

## Expense Approval Workflow

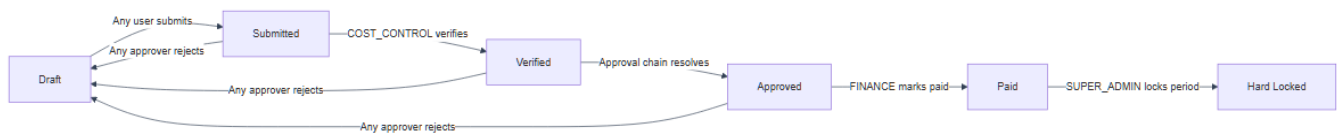


Figure 2

### Approval Chain Resolution

The approval chain is built at submit time from the `approval_rules` table based on:

- Expense **amount** (range matching)
- Expense **cost code category** (Direct, Site, Personnel, Overhead, Other)

Default example matrix:

| Amount Range            | Required Approvers (in order)    |
|-------------------------|----------------------------------|
| IDR 0 – 50,000          | COST_CONTROL                     |
| IDR 50,001 – 500,000    | COST_CONTROL → PM                |
| IDR 500,001 – 2,000,000 | COST_CONTROL → PM → FINANCE      |
| IDR 2,000,001+          | COST_CONTROL → PM → FINANCE → MD |

The chain is stored as JSON on the expense record and is **immutable** once submitted.

## Legal Document Workflow

*Figure 3***Document types:**

- proposal — Surat Penawaran (SPH/YYYY/###)
- berita\_acara — Completion Certificate (BA/YYYY/###)
- surat\_jalan — Delivery Order (SJ/YYYY/###)
- other — General Letter (SRT/YYYY/###)

## Menu-Based Access Control

In addition to role-based guards, each user has a **per-menu permission toggle** (UserMenuPermission). SUPER\_ADMIN can grant or revoke individual menu access per user, overriding the role defaults. This allows fine-grained control (e.g. giving a specific STAFF member access to the Reports page).